

Q1 - 24 January - Shift 1

A hole is drilled in a metal sheet. At 27°C , the diameter of hole is 5 cm. When the sheet is heated to 177°C , the change in the diameter of hole is $d \times 10^{-3}$ cm. The value of d will be _____ if coefficient of linear expansion of the metal is $1.6 \times 10^{-5}/^{\circ}\text{C}$.

Space for your notes:

Q2 - 30 January - Shift 2

A faulty thermometer reads 5°C in melting ice and 95°C in steam. The correct temperature on absolute scale will be..... K when the faulty thermometer reads 41°C .

Space for your notes:

Q3 - 31 January - Shift 2

A water heater of power 2000 W is used to heat water. The specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$. The efficiency of heater is 70%. Time required to heat 2 kg of water from 10°C to 60°C is _____ s.

Space for your notes:

(Assume that the specific heat capacity of water remains constant over the temperature range of the water).

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (12)

Q2 (313)

Q3 (300)

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Q1 (12)

 d_0 at 27°C & d_1 at 177°C

$$d_1 = d_0 (1 + \alpha \Delta T)$$

$$d_1 - d_0 = 5 \times 1.6 \times 10^{-5} \times 150 \text{ cm}$$

$$= 12 \times 10^{-3} \text{ cm}$$

Q2 (313)

$$\frac{41^\circ - 5^\circ}{95^\circ - 5^\circ} = \frac{C - 0^\circ}{100^\circ - 0^\circ}$$

$$\Rightarrow C = \frac{36}{90} \times 100 = 40^\circ\text{C} = 313 \text{ K}$$

Q3 (300)

No Solution